

Make Your Data *Talk*

Five plots, one style baseline, zero gray backgrounds — work the Titanic dataset until your charts read like a decision, not a default.

NAME

DATE

DATASET USED

1

Kill the Gray Background

- Load `sns.load_dataset('titanic')`. Plot `df['fare'].hist()` raw — **zero styling**. Screenshot it.
- Run `sns.set_theme(style='whitegrid', palette='muted', font_scale=1.2)` plus the `rcParams` block (dpi 150, figsize 10x6, spines off). Replot same histogram.
- Put both side by side. Note **3 differences** you see.

```
sns.set_theme(style="whitegrid", palette="muted", font_scale=1.2)
```

3 DIFFERENCES SPOTTED

2

Distribution Skew Check

- Run `sns.histplot(kde=True)` on **both** `df['age']` and `df['fare']` side by side using `subplots`.
- Fare skews right — most under **£50**, some over *£500*.
- Write one sentence: raw fare vs log -transformed fare as a model feature — which wins, why?

RAW VS LOG-TRANSFORM VERDICT

3

Encode a Third Variable

- Build age-vs-fare scatterplot: `hue='survived', colors={0:'salmon',1:'steelblue'}`.
- Swap to `hue='pclass'` — same axes, same data.
- Which hue reveals a **clearer pattern**? Explain in *2 sentences*.

CLEARER HUE + WHY

4

Read the Confidence Interval

- Reproduce `sns.barplot(x='pclass', y='survived', errorbar='ci')`. Log **rate + CI width** per class.
- Rerun grouped by `'sex'` instead.
- CI tighter or wider? What does that say about *sample size* per group?

CI WIDTH: PCLASS VS SEX

5

Correlation Heatmap Audit

- Generate `sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm', center=0)`.
- Find **strongest positive** pair (excluding self-correlations).
- Find **strongest negative** pair. Write what each implies about the *Titanic* data.

TOP POS/NEG PAIR + MEANING